Homework 3

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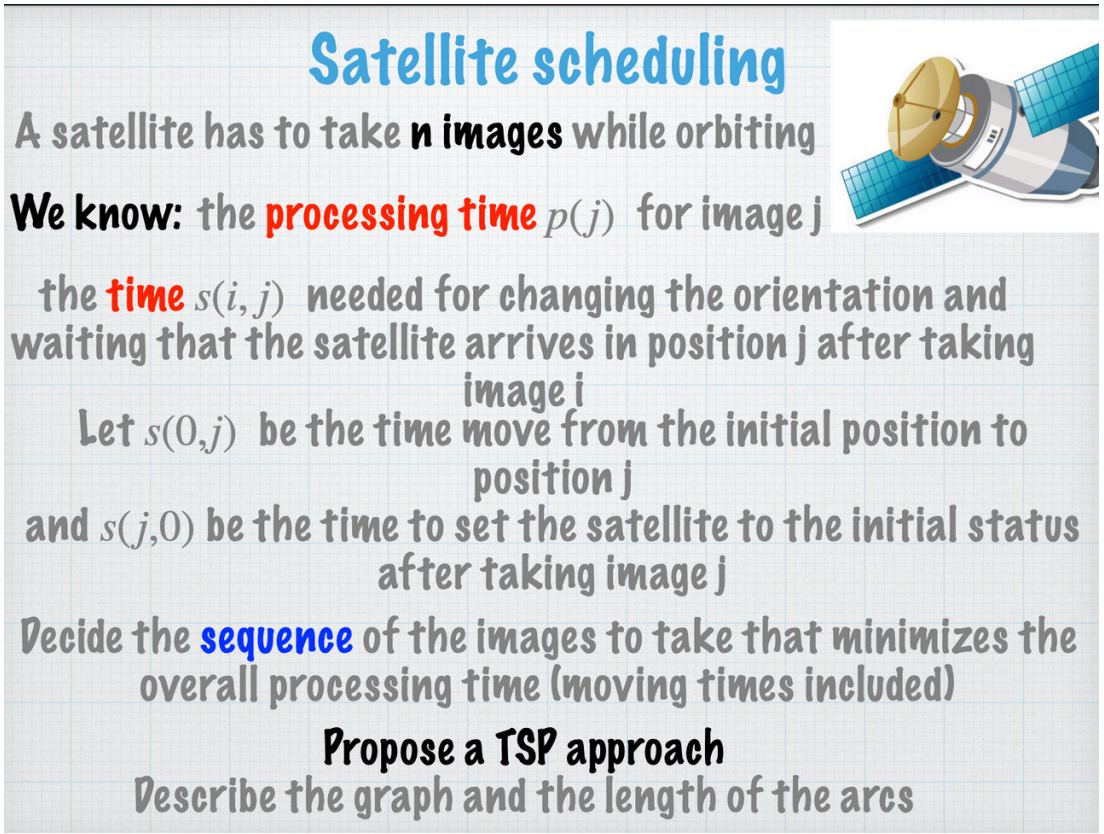
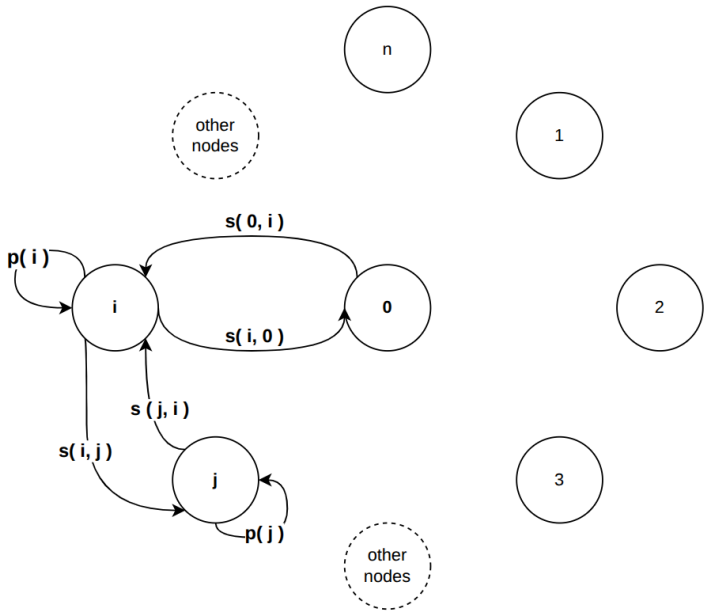
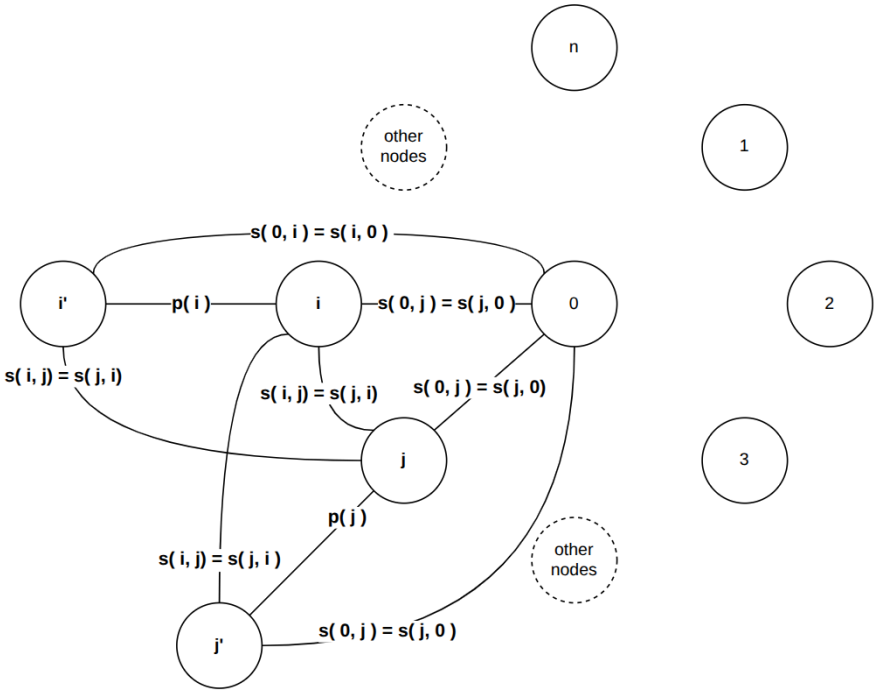


Since the problem is trying to find the better sequence of images to take in order to minimize the overall time consumed, a TSP approach should model the different spots where photos are to be taken as different **nodes** distributed in the graph space, the "*movements*" of the satellite as **arcs** going from nodes to others and the time consumed into taking a photo as an arc that points from a node to **itself**. By adding the initial position `0` and omitting the other arcs for better visibility, we have a graph that looks like this:



If we want to model the problem following a **TSP** approach it is necessary to make the graph **undirected**: it is reasonable that the time of arrival from position `i` to position `j` is the same if start and finish position are swapped, but we need to change the fact that the action of taking a photo is modeled with a self referring arrow. We must introduce a new state for each position where we can take a picture to correctly model the actual cost of taking it:



With this graph, once a photo is taken (`x'` nodes) it is possible to go to the next position (`x` nodes) by paying the cost of travel, it is also possible to take multiple pictures in the same spot (even if its not efficient) and its guaranteed that, by modeling the action of taking a picture as a node, we must take every picture of every location to solve the TSP-like problem. Now it is possible to apply the usual algorithms of optimization to try and find the best path (being the problem NP-complete).